

Pose questions to explore observed patterns and relationships and make predictions based on observations

Learning intention

- pose questions to explore observed patterns and relationships and make predictions based on observations.

Teach and model

- predicting which material will be the most effective insulator of heat.
- predicting whether the mass of ice in a sealed container will change when the ice has melted.
- predicting how quickly ice will melt at different ambient temperatures based on previous observations.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.